Increasing purchase events

Conversion opportunities and purchase follow-up strategies

18 November–18 December 2014

Source: user-provided SQL results and updated deck. Observation period: 18 Nov–18 Dec 2014.

PURCHASE BASELINE

Grow purchase events through conversion and additional purchases

120,205

Recorded purchase events

8,886

Unique buyers across all categories

Business objective

Increase total purchase events by improving engage-to-buy conversion, encouraging additional purchases from repeat-purchasing buyers, and recovering unpurchased carts.

All categories: 81,659 purchasing shopper-category combinations.

Source: user-provided SQL results and updated deck. Observation period: 18 Nov–18 Dec 2014.

Three methods to test for additional purchase events

01 Use surveys to diagnose conversion barriers

Survey engaged shoppers who did not purchase in large-audience, below-average-conversion categories. Investigate purchase intent, availability, price, shipping costs, and product information; use the findings to select targeted improvements.

02 Encourage repeat purchasing

Repeat purchasing is one potential way to generate additional purchase events from existing buyers. Test subscriptions where replenishment needs are confirmed; recommend relevant complementary products in lower-repeat-purchase categories.

03 Follow up on unpurchased carts

For shoppers who have not purchased in the category after its P80 waiting period, test cart reminders to encourage purchase.

Success: more purchase events than the control group, with equal group sizes and follow-up periods.

Source: user-provided SQL results and updated deck. Observation period: 18 Nov–18 Dec 2014.

Narrowing the analysis from 8,916 total categories to 680 selected categories

12.3M
Behavior events

10K
Unique users

8,916
Item categories

Definition	Analysis rule
Selected categories	At least 30 click users and 30 engaged shoppers per category
Engage	A save or add-to-cart event strictly after the first click
Unit of analysis	Shopper-category combination. Each shopper may appear in multiple categories and is counted once per category.
Purchase measure	Recorded buy events, not orders or revenue

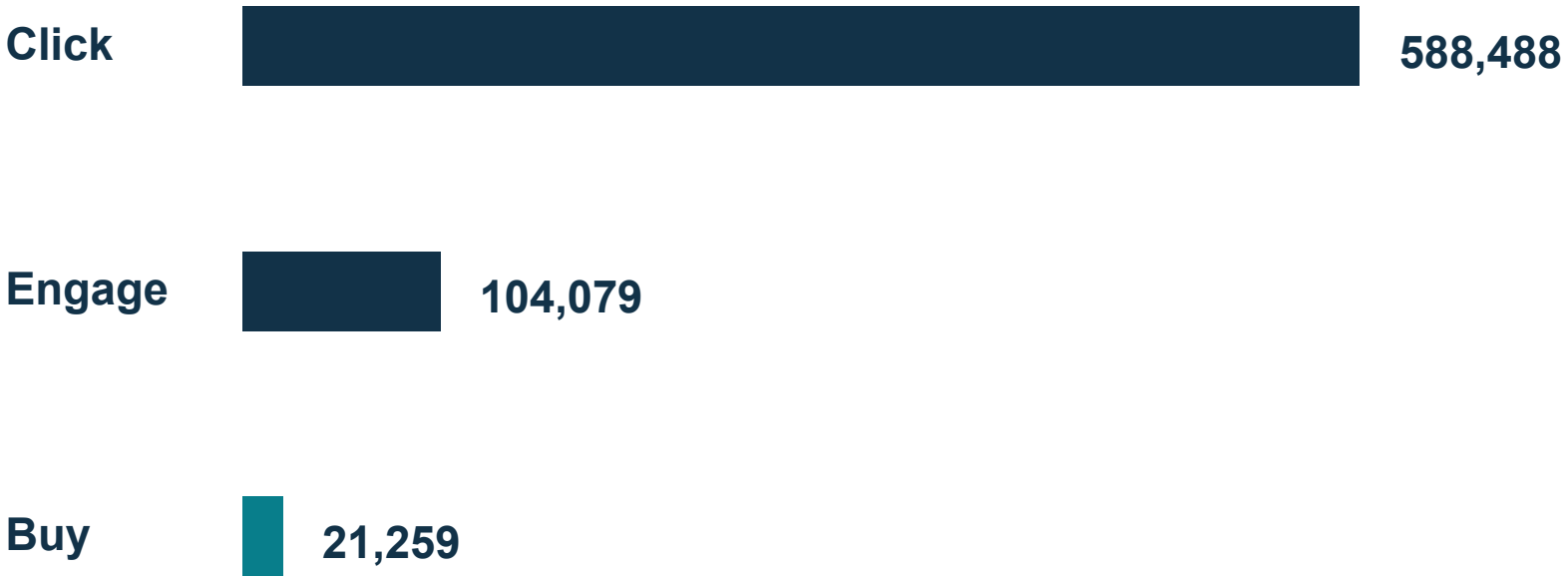
One shopper-category combination may generate multiple purchase events.

Source: updated user deck. 2,876,947 unique items; timestamps recorded to the hour. Observation: 18 Nov–18 Dec 2014.

CONVERSION FUNNEL

20.43% of engaged shoppers purchased in the same category

Shoppers counted once per category at each stage



17.69%

Click-to-engage

20.43%

Engage-to-buy

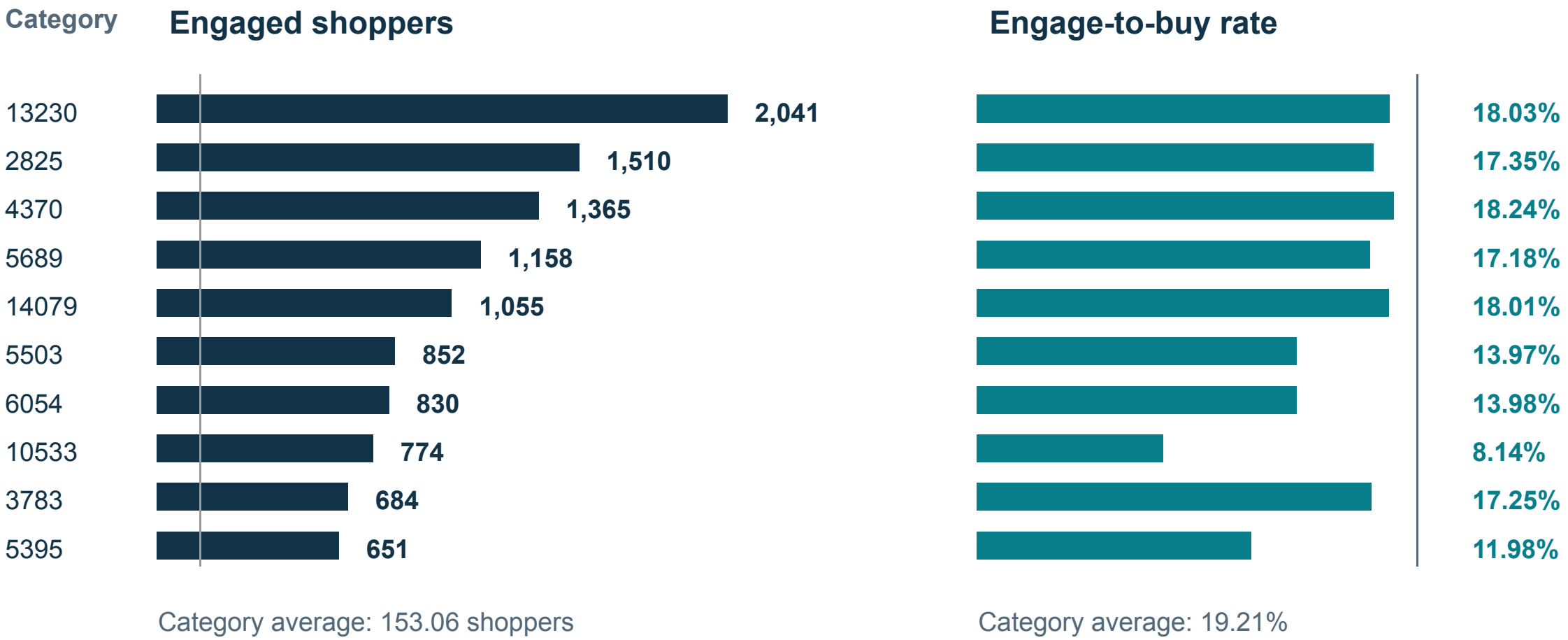
3.61%

Full-funnel conversion

Buy stage: 21,259 purchasing shopper-category combinations

680 selected categories. Each stage counts shopper-category combinations once. Strictly later events within the same category.

Prioritize categories with high engagement and low conversion



Selection: engaged shoppers >153.06 and conversion <19.21%; top 10 by engaged shoppers. Category means differ from the pooled 20.43% rate.

Use surveys to diagnose conversion barriers

CATEGORY 13230

2,041 engaged shoppers
18.03% engage-to-buy rate

Survey a large engaged audience

Survey engaged shoppers who did not purchase to identify common barriers across this large audience.

CATEGORY 10533

774 engaged shoppers
8.14% engage-to-buy rate

Investigate the largest conversion gap

Ask engaged shoppers who did not purchase about purchase intent, availability, price, shipping costs, and product information.

Use survey findings to select and test targeted improvements.

Reference: category averages of 153.06 engaged shoppers and a 19.21% engage-to-buy rate.

Repeat purchase patterns guide post-purchase offers

12.70% overall repeat purchase rate = 2,700 repeat-purchasing ÷ 21,259 purchasing shopper-category combinations

2,700 purchased on at least two distinct dates; 21,259 purchased at least once within the funnel.
Because purchase events do not necessarily represent separate orders, repeat purchase requires at least two distinct dates within the same category.

Higher-repeat-purchase categories ≥12.70%

Category	Records	Repeat rate
1863	813	19.43%
5232	475	22.11%
6513	424	16.51%
5399	405	20.49%
5027	402	17.16%
5894	399	18.55%
13230	368	19.02%
10894	322	19.25%
6344	303	17.82%
2825	262	16.79%

Lower-repeat-purchase categories <12.70%

Category	Records	Repeat rate
11279	316	11.08%
11824	180	11.67%
3628	173	10.98%
11552	163	11.66%
3381	161	9.94%
6648	141	0.71%
2513	134	10.45%
3472	134	7.46%
5503	119	12.61%
6054	116	12.07%

Use category-level patterns to tailor offers and generate additional purchase events.

Funnel only; purchases after engage. At least 30 purchasing shopper-category combinations per category; top 10 per group by count.

METHOD 2 / SOLUTION

Tailor post-purchase offers to category repeat behavior

HIGHER-REPEAT-PURCHASE CATEGORIES

Repeat purchase rate ≥12.70%

Test subscription offers

Offer subscriptions where replenishment needs are confirmed. Test repeat-purchase discounts as an alternative.

LOWER-REPEAT-PURCHASE CATEGORIES

Repeat purchase rate <12.70%

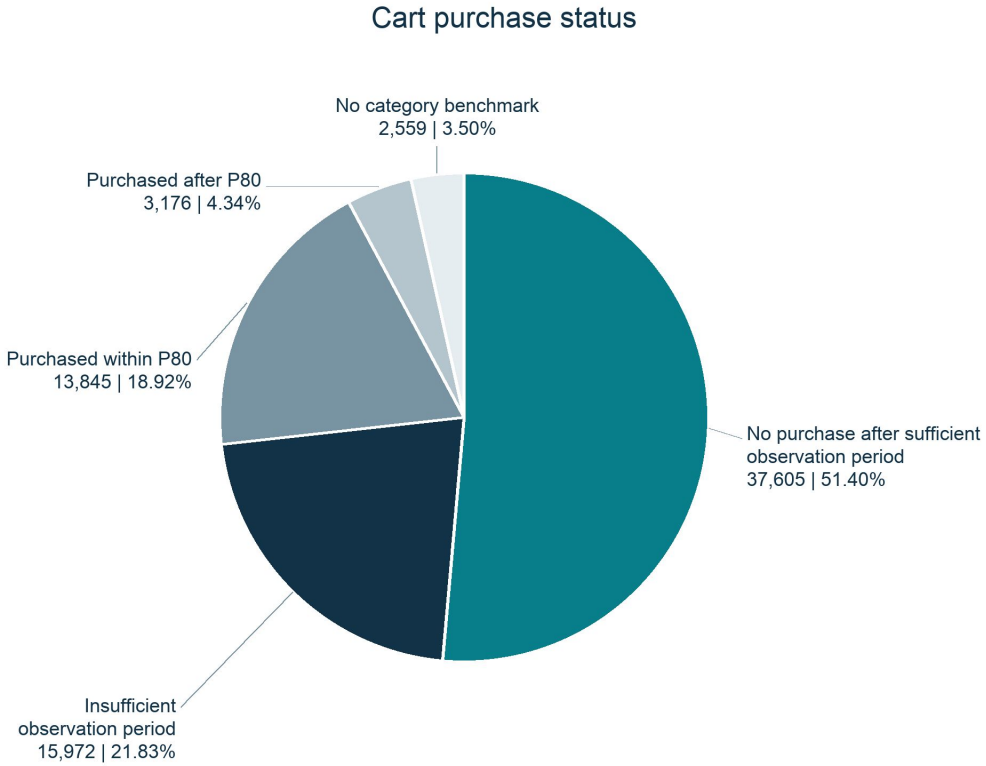
Recommend complementary products

Recommend relevant complementary products after purchase, after verifying product compatibility and relevance.

Measure additional purchase events across all categories.

Include complementary purchases in the outcome. Repeat behavior alone does not establish subscription suitability.

37,605 shopper-category combinations have an add-to-cart event but no subsequent purchase



Status	Records	Share
No purchase after sufficient observation period	37,605	51.40%
Insufficient observation period	15,972	21.83%
Purchased within P80	13,845	18.92%
Purchased after P80	3,176	4.34%
No category benchmark	2,559	3.50%

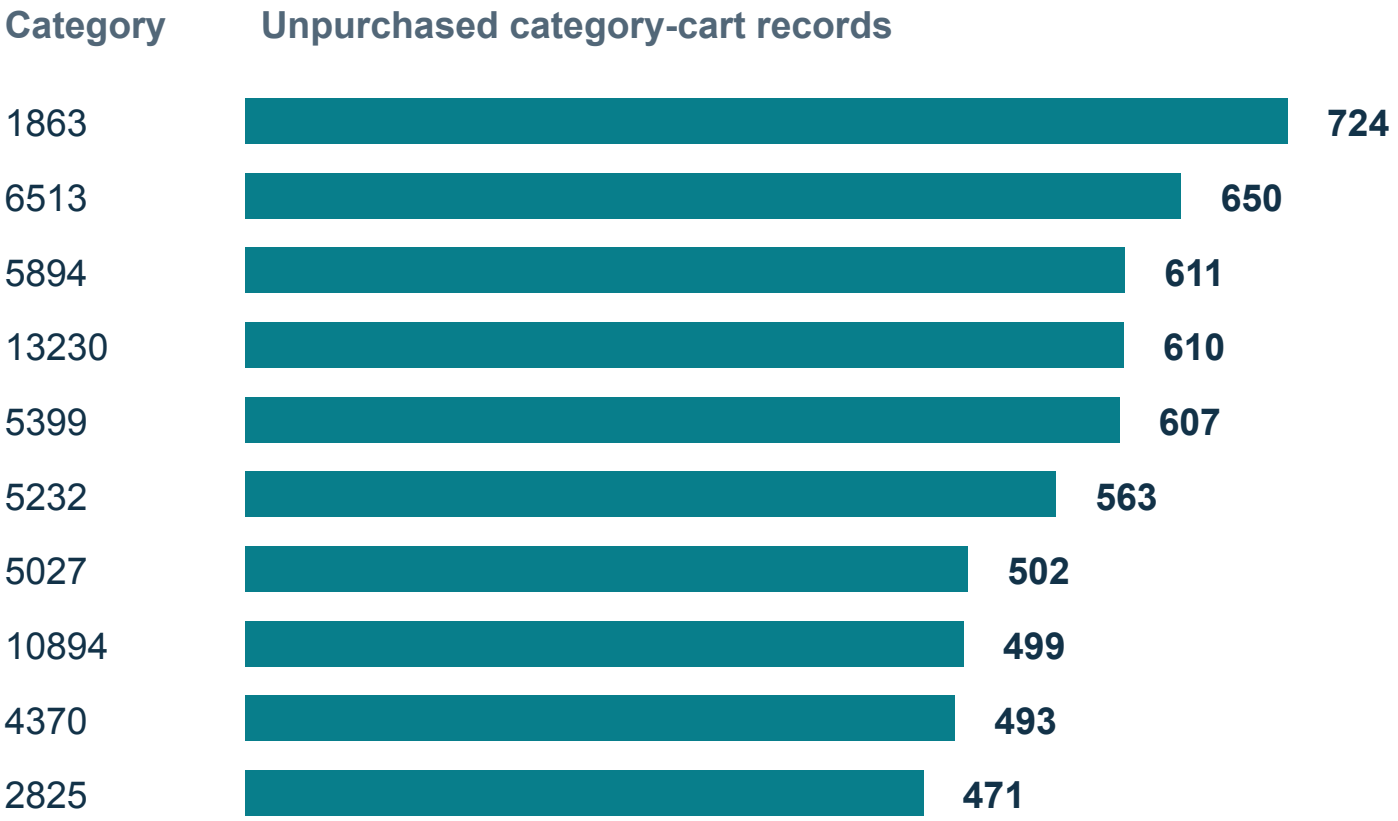
P80: 80th percentile of observed cart-to-buy times within each category.

Reminder test audience

Recheck purchase status before sending. P80 identifies test candidates, not confirmed abandonment.

Unit: shopper-category combinations with an add-to-cart event. First cart after first click; no subsequent purchase beyond P80.

The top 10 categories account for 15.24% of shopper-category combinations with an add-to-cart event but no subsequent purchase



5,730

Shopper-category combinations with an add-to-cart event but no subsequent purchase

Includes only records with no purchase observed after the category's P80 waiting period. Ranked by record count.

METHOD 3 / SOLUTION

Test timely reminders for shoppers who have not purchased

01 / WAIT

Wait until category P80

Use the category's observed cart-to-buy waiting period to identify follow-up candidates.

02 / CHECK

Recheck purchase status

Include only shoppers who still have no purchase in the category before sending a reminder.

03 / TEST

Test a cart reminder

Randomly assign eligible users to reminder and no-reminder groups.

Compare purchase events over the same follow-up period.

P80 is a test trigger based on observed purchasing times, not proof of cart abandonment.

The findings identify test candidates, not causal uplift

Short observation period	The data covers only one month, so later purchases may be missed, especially for shoppers who engaged near the end of the period.
Measurement and scope	The analysis counts purchase events rather than orders and focuses on 680 selected categories. Shoppers may be counted in more than one category.
Category comparisons	Purchase cycles vary by category. A lower repeat rate may reflect less frequent purchase needs rather than a problem with customer retention.
Cart recovery threshold	P80 is based only on shoppers who purchased, sometimes with as few as five observations. Shoppers who exceed this threshold may still purchase later.
Missing product information	Category IDs alone do not show which products suit subscriptions or complement each other. These recommendations require product-level validation.

Source: user-provided SQL results and updated deck. Observation period: 18 Nov–18 Dec 2014.

Three actions to test for additional purchase events

01	**Use surveys to diagnose conversion barriers**	Survey engaged shoppers who did not purchase to identify barriers, then use the findings to select and test targeted improvements.
02	**Tailor post-purchase offers**	Test subscriptions or repeat-purchase discounts in higher-repeat-purchase categories, and relevant complementary-product recommendations in lower-repeat-purchase categories.
03	**Test cart reminders**	Test cart reminders for shoppers who have not purchased after the category's P80 waiting period.

Success: more purchase events than the control group.
Test setup: equal group sizes and follow-up periods. Run pilots separately or use mutually exclusive test groups.

Source: user-provided SQL results and updated deck. Observation period: 18 Nov–18 Dec 2014.